

crewai

Conceitos de Crew

Do Básico ao Avançado

Versões

crewai[tools]>=1.7.0

Python 3.12.9

AI Agents?

LLMs

```
graph TD; LLMs[LLMs] --> Content[Incrível em criar conteúdo]; Content --> Reasoning[Tem raciocínio  
(Pode fazer tomar decisões sensatas)];
```

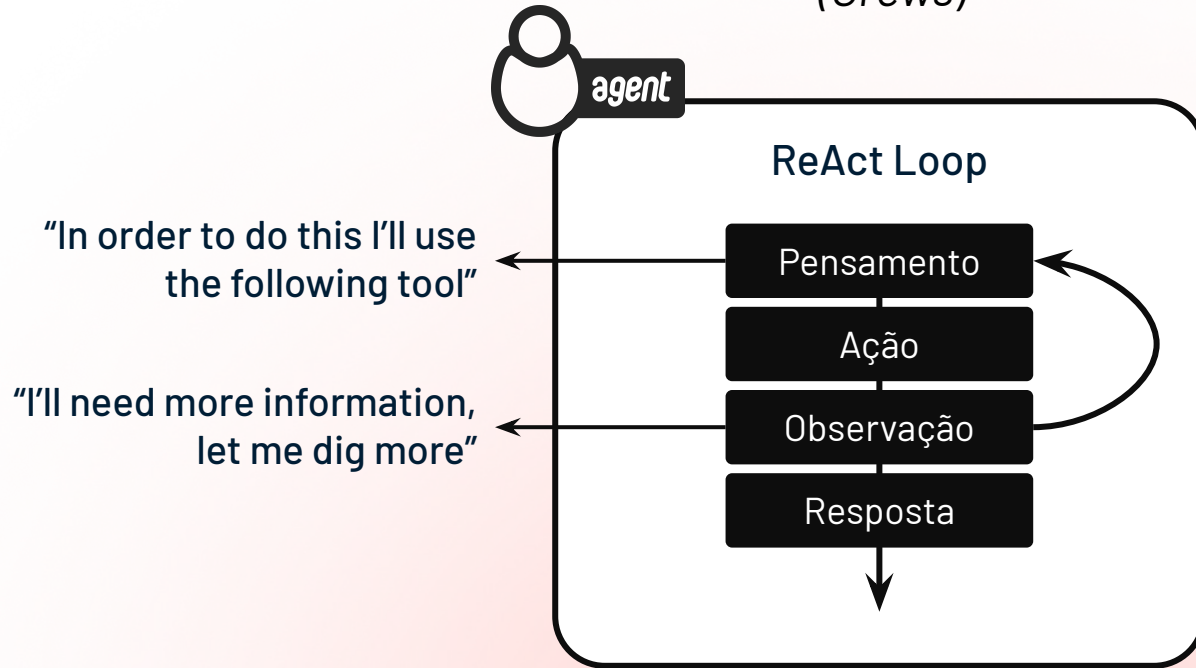
**Incrível em criar
conteúdo**

Tem raciocínio
*(Pode fazer tomar decisões
sensatas)*

**AI Control the
Application Flow**

Agent Workflow

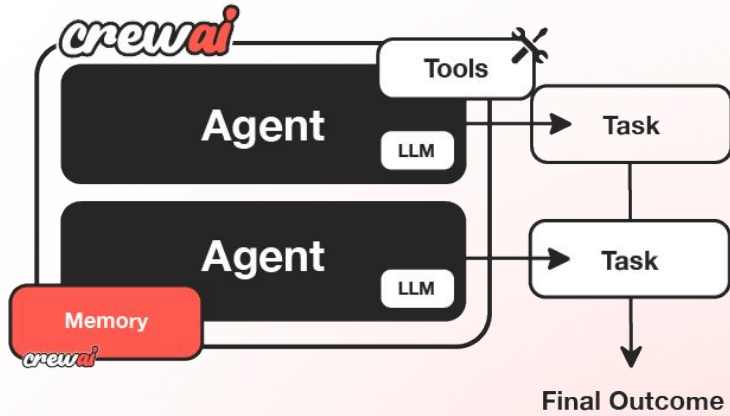
(Crews)



crewai create crew <nome-da-crew>

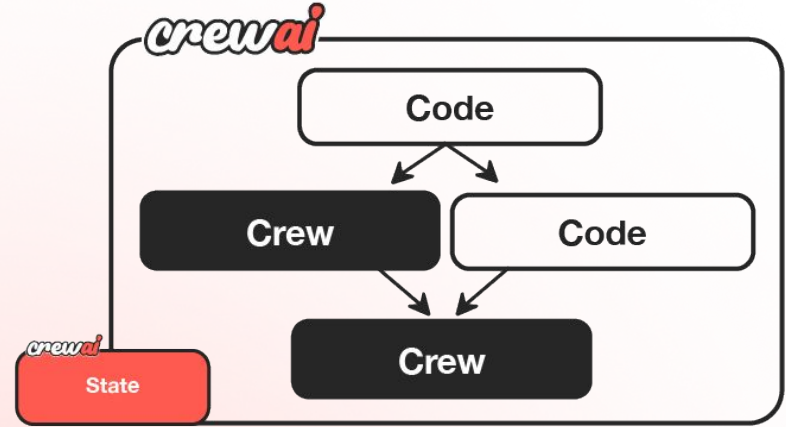
Crew

more agency



Flows

finer precision



Agent Definition

Think like a manager!

Designing Agent Role (Job Title)

- **Específico & especializado** (ex: “Clinical Trial Data Auditor”)
- De acordo com **papéis da vida real**
- Expressa especialidade no domínio

```
# Create specialized agents following the Role-Goal-Backstory framework
financial_analyst = Agent(
    role="Financial Data Analyst specializing in SaaS metrics",
    goal=f"Extract and analyze key financial data points for {year} with a focus on accuracy and context",
    backstory=f""""With 12 years of experience in financial analysis at top investment firms,
you've developed a methodical approach to financial data extraction and normalization.
You focus on SaaS-specific metrics and have a talent for identifying early warning signs
in financial statements. Your analysis always includes both GAAP and non-GAAP metrics
with a particular emphasis on cash flow and unit economics. You're analyzing data for the year {year}.""",
    verbose=True,
    allow_delegation=False,
    tools=[serper_tool, scrape_tool],
    llm=llm
)
```

Agent Goal (Brief Expectations)

- Focado e **mensurado** no *Resultado*
- Incorporar **padrões de qualidade** E **critérios de sucesso**
- O que seria “**bom**” ?

```
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    verbose=True,  
    allow_delegation=False,  
    tools=[serper_tool, scrape_tool],  
    llm=llm  
)
```

Agent Backstory (Job Experience)

- Declare **habilidades** e **experiência**
- Descreva estilo de trabalho e valores
- Garanta consistência com **Papel** e **Objetivo** (role & goal)

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    verbose=True,
    allow_delegation=False,
    tools=[serper_tool, scrape_tool],
    llm=llm
)
```

Especialistas > Generalistas

Agentes especialistas entregam `outputs` precisos

Example:

Technical Blog Writer Agent ao invés de **Writer Agent**

Se o objetivo é ter um agente de criação de conteúdos de blog.

Balancear especialidade e Versatilidade

- Especialize no papel, versátil na prática
- Evite escopos excessivamente restritos
- Habilidades complementares na Crew

Prompting

Agent

Task

Role
Goal
Backstory

Description
Expected Output



Melhores práticas: A Regra 80/20

- **80% esforço:**
 - Elabore tarefas claras (descrição da tarefa e resultado esperado).
- **20% esforço:**
 - Aprimorar personas de Agentes

“Well-designed tasks rescue imperfect agents”

```
# Following the 80/20 rule - focusing on detailed task design
financial_analysis_task = Task(
    description=f"""
    Analyze {company_name}'s ({ticker}) financial performance for {year} focusing on:
    1. Revenue growth (YoY and 5-year CAGR)
    2. Profit margin trends (last 8 quarters)
    3. Cash flow stability (operating and free cash flow)
    4. SaaS-specific metrics (ARR, CAC, LTV, churn)
    5. Balance sheet health and liquidity

    Use the SerperDevTool to search for real financial data for {ticker}.
    Search for specific queries like "{ticker} financial statements {year}",
    "{ticker} annual report {year}", and "{ticker} investor relations earnings".

    Use the ScrapeWebsiteTool to extract detailed data from financial websites.
    Try scraping from sites like:
    - The company's investor relations page (search for "{company_name} investor relations")
    - SEC EDGAR database (use css selectors to extract financial tables)
    - Financial news sites with analysis of {company_name}

    Your analysis should incorporate real data from these tools, comparing key metrics
    to SaaS industry benchmarks where applicable. Use a first-principles approach to
    evaluate the financial health of the business.

    Focus on data and performance specific to the year {year}.

    Important: Format your analysis in markdown with clear sections and bullet points.
    """,
    expected_output=f"""Financial Analysis for {company_name} ({ticker}) for {year}""",
    agent=financial_analyst
)
```


A regra 80/20: Prefira focar na Tarefa ao invés dos Agentes

Ao construir sistemas de IA eficazes, lembre-se deste princípio crucial: 80% do seu esforço deve ser dedicado ao planejamento das tarefas.

Mesmo o agente mais perfeitamente definido falhará com tarefas mal projetadas, mas tarefas bem projetadas podem elevar o desempenho até mesmo de um agente simples. Isso significa:

Por que o design do agente é importante

Saída de qualidade

Agentes bem projetados produzem resultados mais relevantes e de alta qualidade.

Eficácia da colaboração

Agentes com habilidades complementares trabalham juntos com mais eficiência.

Performance de Tarefas

Agentes com funções e objetivos claros executam tarefas com mais eficácia.

Sistema escalável

Agentes cuidadosamente projetados podem ser reutilizados em diversas equipes e contextos.



Agents Repository

Manage your agents - Create and configure agents to automate tasks and workflows ⓘ

+ Add Agent



Agent Repository

1. Create agents with specific roles and goals for your workflows
2. Configure tools and capabilities for each specialized assistant
3. Deploy agents across projects via visual interface or API integration



Your Pre Made Agents

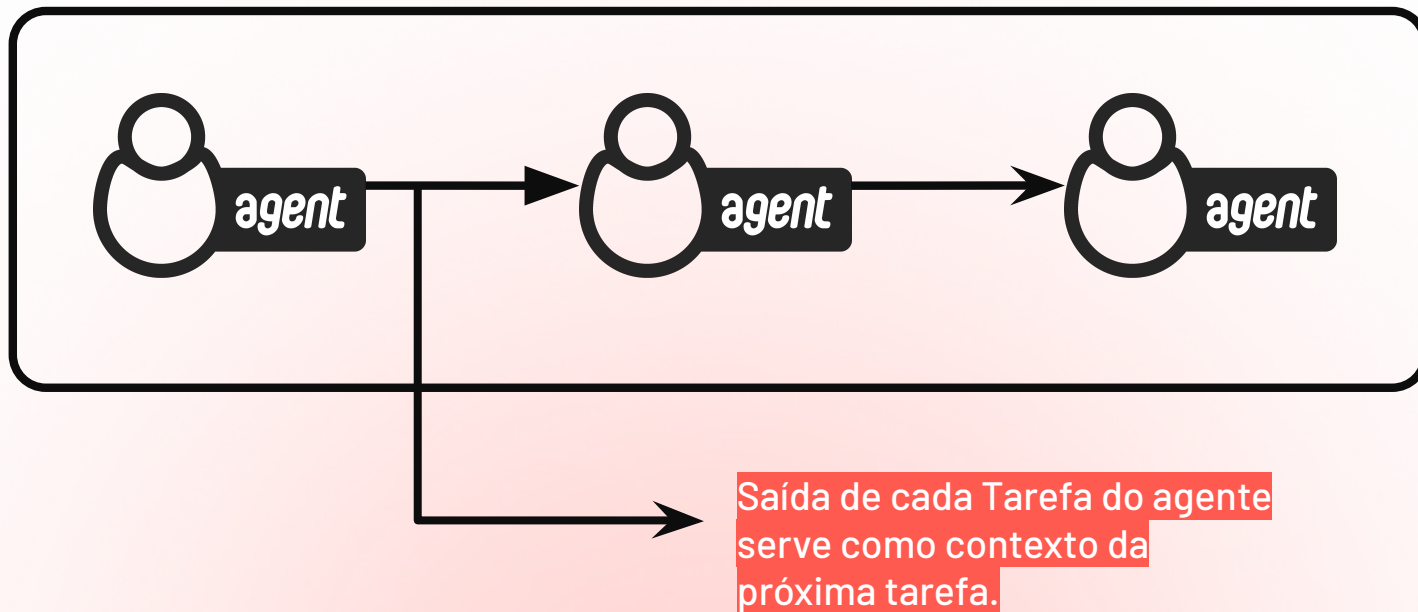
ROLE	GOAL	TOOLS			ACTIONS
Sales Research	gather sales data about a lead	crewai-tools: ScrapeWebsiteTool	crewai-tools: SerperDevTool	crewai-tools: CrewaiEnterpriseTools	Options ▾ Delete

Agent Collaboration

Sequential vs Hierarchical

Sequential Process

Crew




```
from crewai import Agent, Task

# Agents get created first
# ...

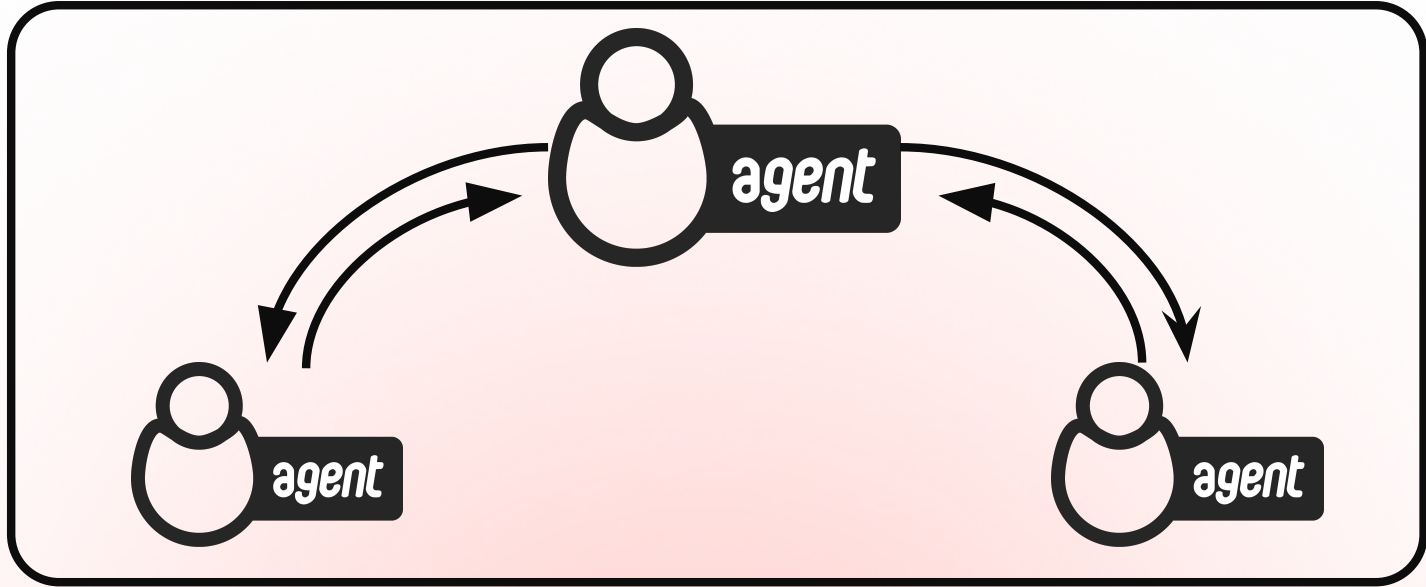
# Create a research task
research = Task(
    description="Research the...",
    expected_output="A detailed...",
    agent=research_agent,
)

# Create a research task
research 2 = Task(
    description="Research the...",
    expected_output="A detailed...",
    agent=research_agent,
)
```

```
# Create a report generator task
report = Task(
    description="Generate detailed...",
    expected_output="A detailed
report...",
    agent=report_agent,
    context=[research, research_2]
)
```

Hierarchical Process

Crew



Problemas comuns de coordenação

Redundant work:

Delimite o escopo das tarefas que cada agente executa para evitar sobreposição; comece com processos paralelos e, em seguida, utilize processos hierárquicos quando necessário.

Slow serial chains:

Marque tarefas independentes com **`async_execution=True`** e utilize o ***context*** de uma tarefa posterior para controlá-las.

Agent Workflow

(Crews)

Adding Control on a Probabilistic System

Memory

Dynamically update context engineering to help agents learn and get better over time

Guardrails

Adding either deterministic or probabilistic (LLM as judge) checks on output

Hooks

Execute deterministic code either before or after Agents

Memory

Agentic Memory

(Crews)

Short Term

During Execution can help agents exchange information with each other beyond the context passed along

Long Term

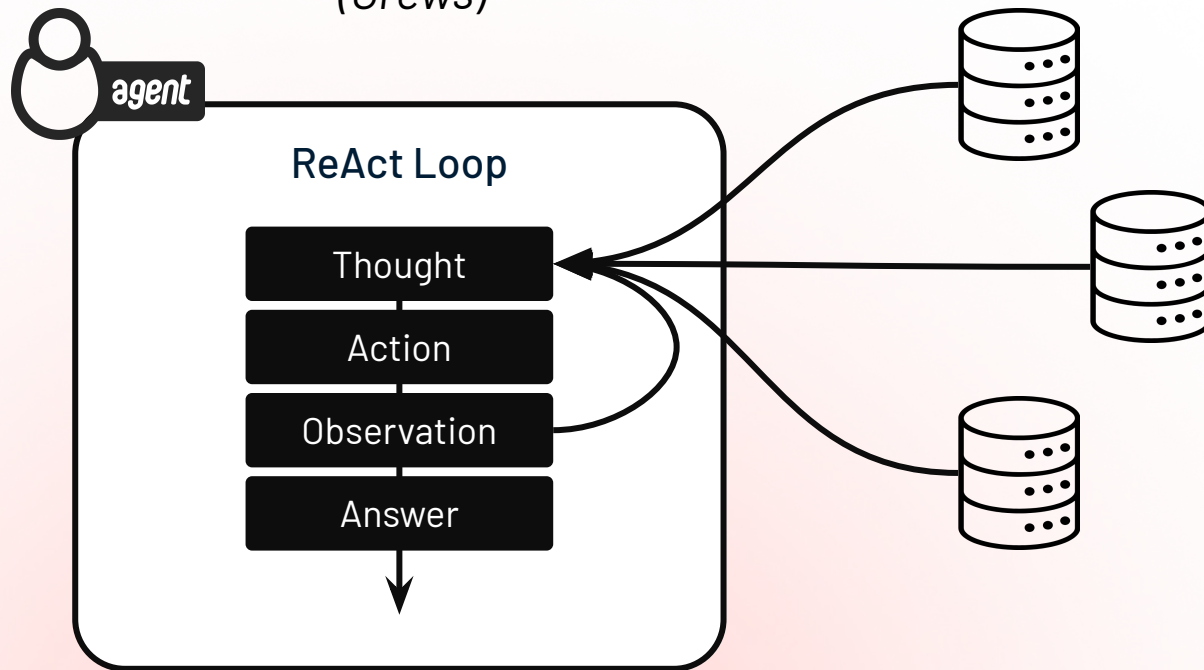
Help agents self-critique and get better over time, understand what it did right and wrong

Entity Memory

Remember "things" names and their definitions, what those mean

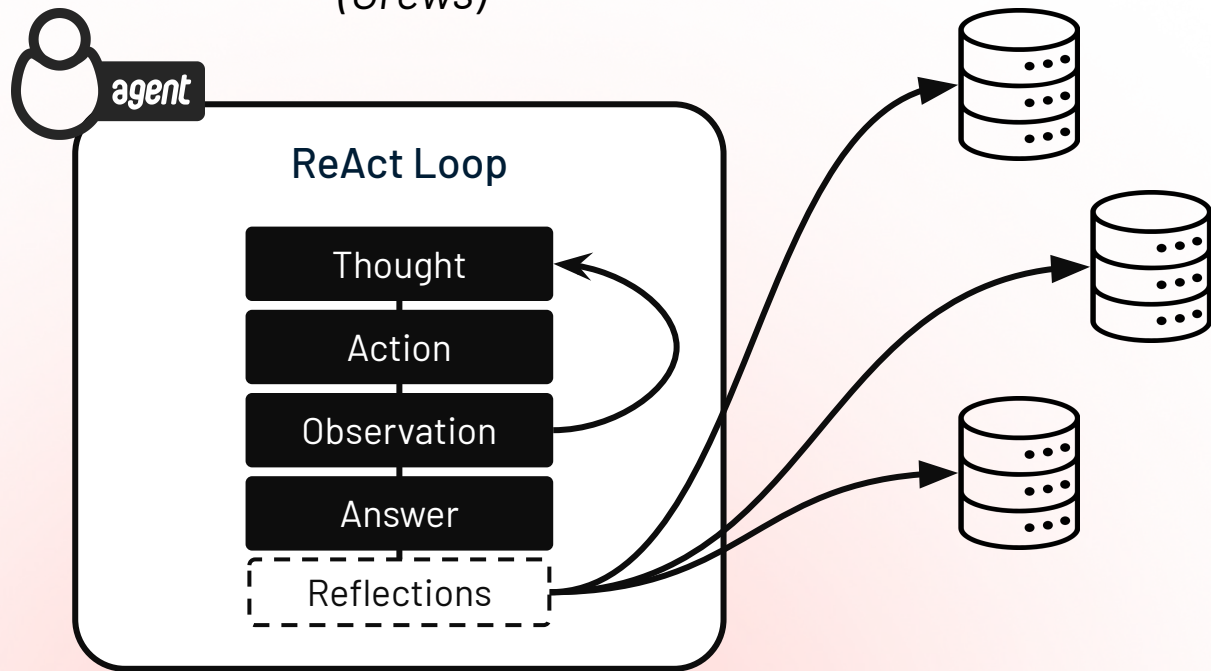
Agentic Memory

(Crews)



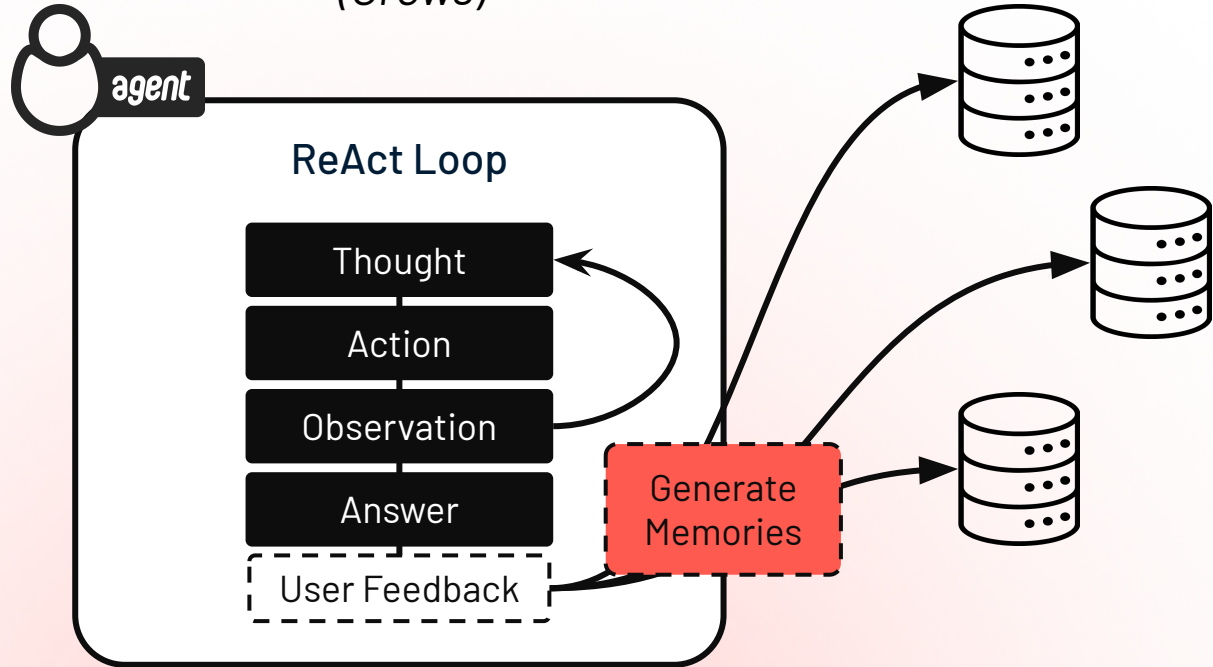
Agentic Memory

(Crews)



Agentic Memory

(Crews)



**Generate
Memories**

**Training with
Feedback**

Agentic Memory

(Crews)

Memory, either by agents or by training is heavily connected to executions and outputs

Extremely helpful for enforcing:

- Behavior
- Format
- Style

Agentic Memory vs Agentic Knowledge

(Crews)

Memory

Data that gets injected into agents executions but generated upon feedback either Human or AI

Knowledge

Data that can be prefilled into a special agent memory, that is used during executions but not impact by it

Knowledge

Agentic Knowledge

(Crews)

Text Sources

Raw Text
Text Files
PDF Documents

Structured Data

CSV Files
Excel Spreadsheets
JSON Documents

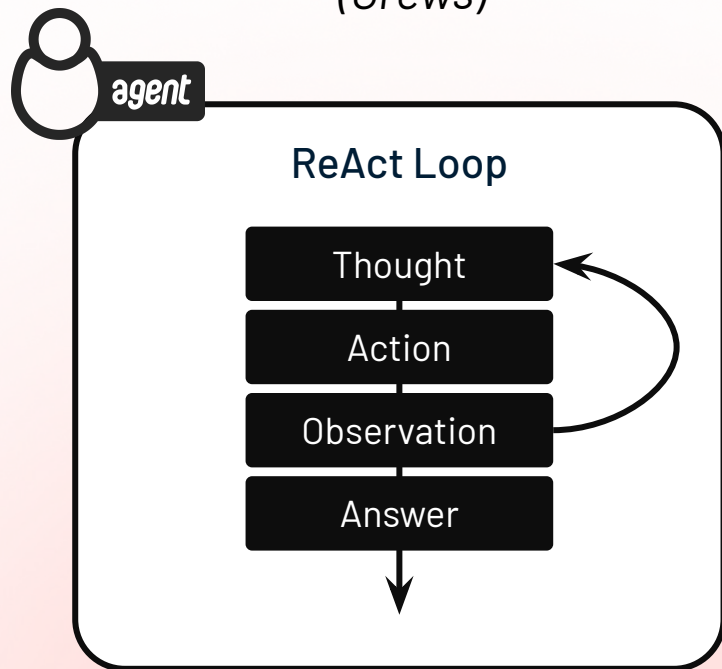
```
from crewai.knowledge.source.text_file_knowledge_source
import TextFileKnowledgeSource
```

```
text_source = TextFileKnowledgeSource(
    file_paths=["document.txt", "another.txt"]
)
```

Guardrails

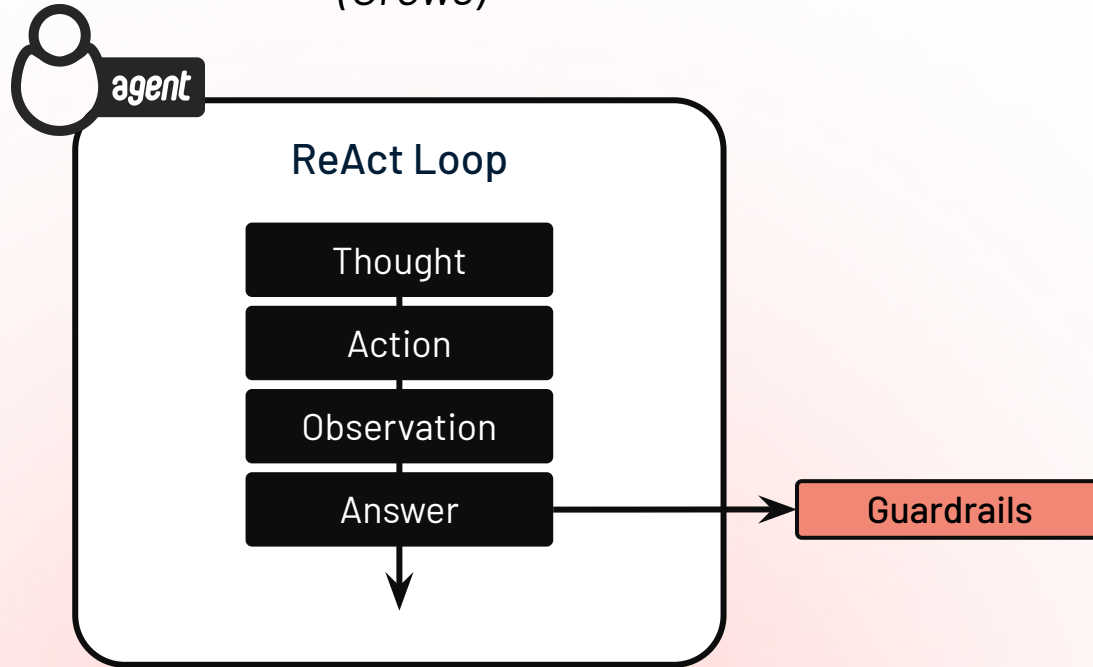
Agent Workflow

(Crews)



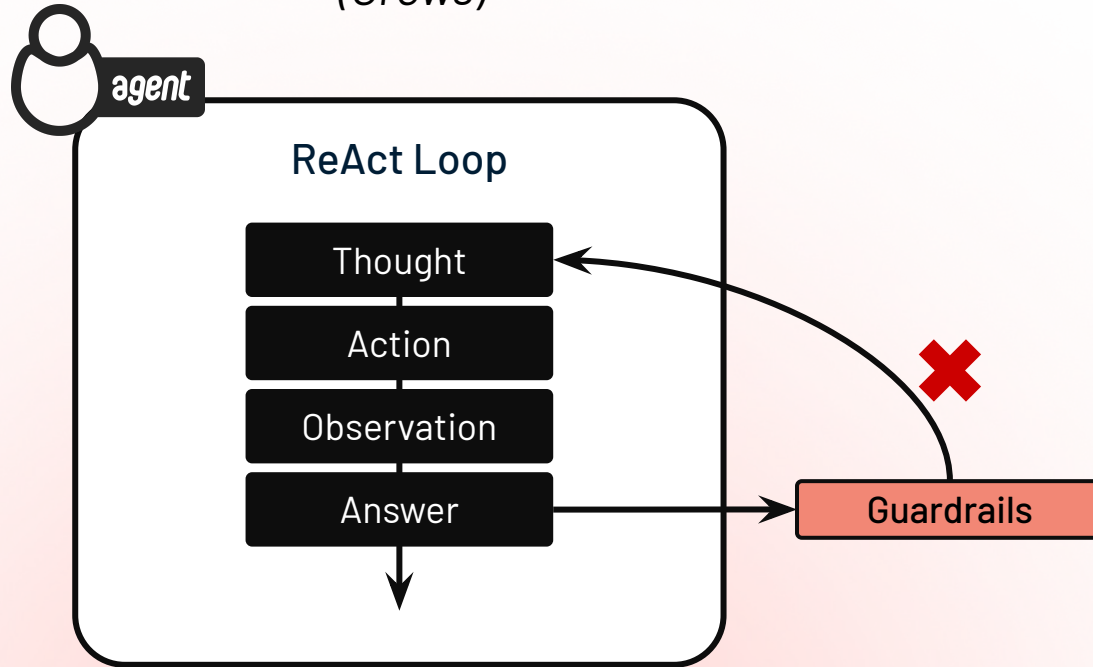
Agent Workflow

(Crews)



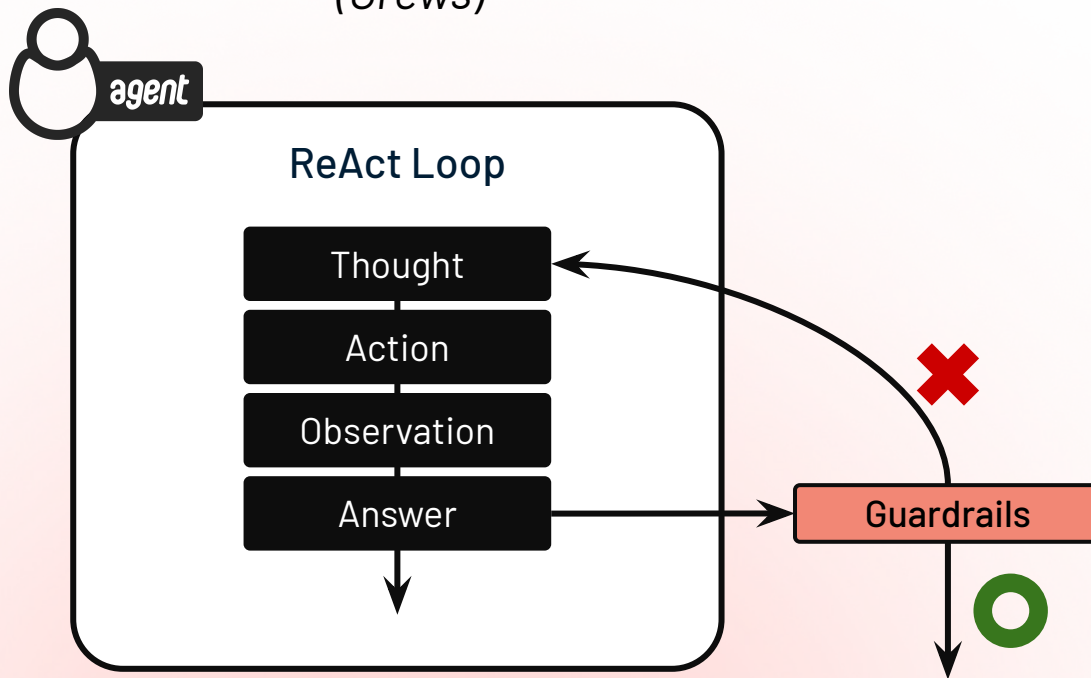
Agent Workflow

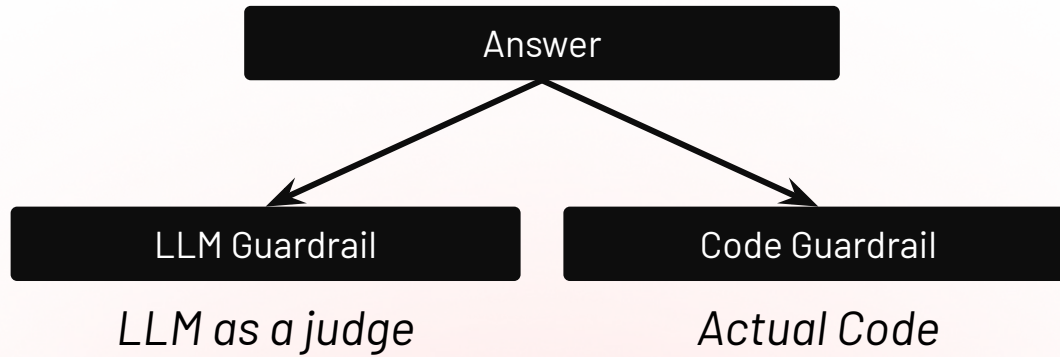
(Crews)



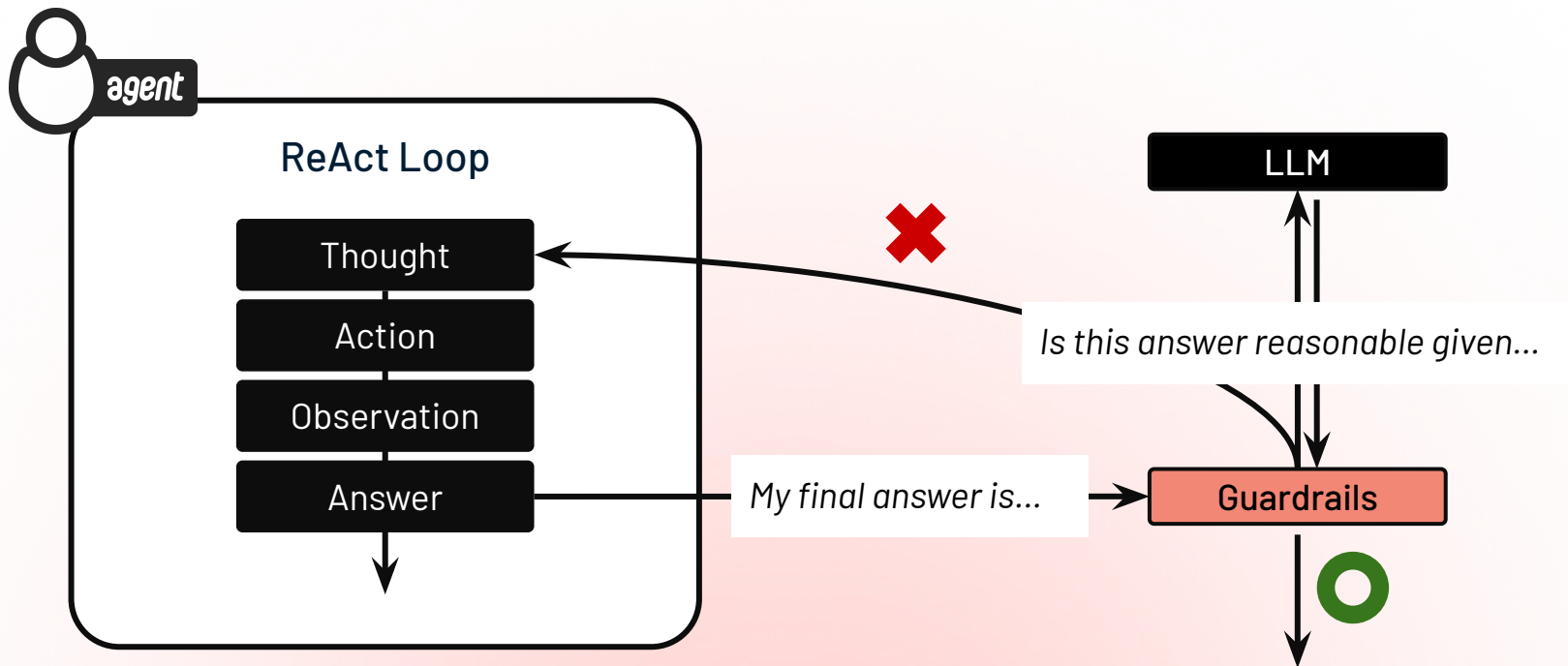
Agent Workflow

(Crews)

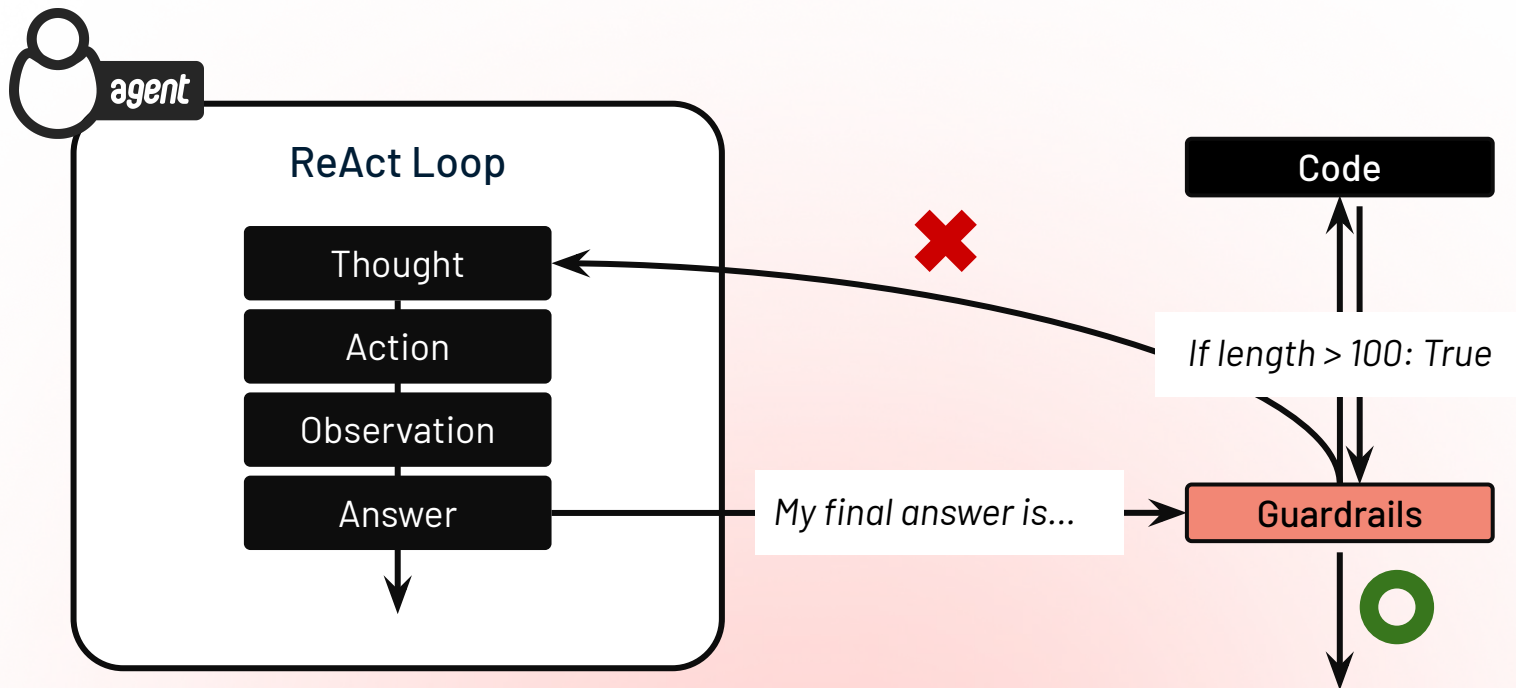




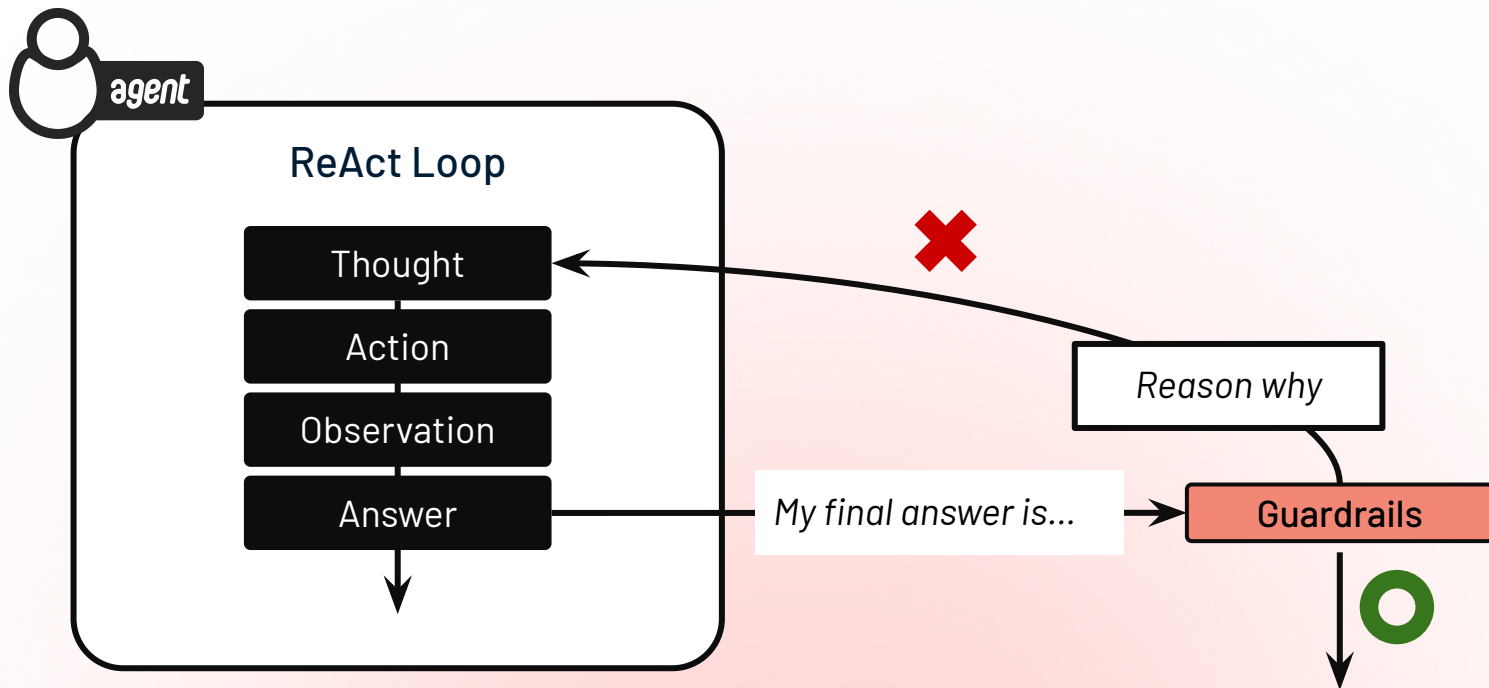
LLM Guardrail



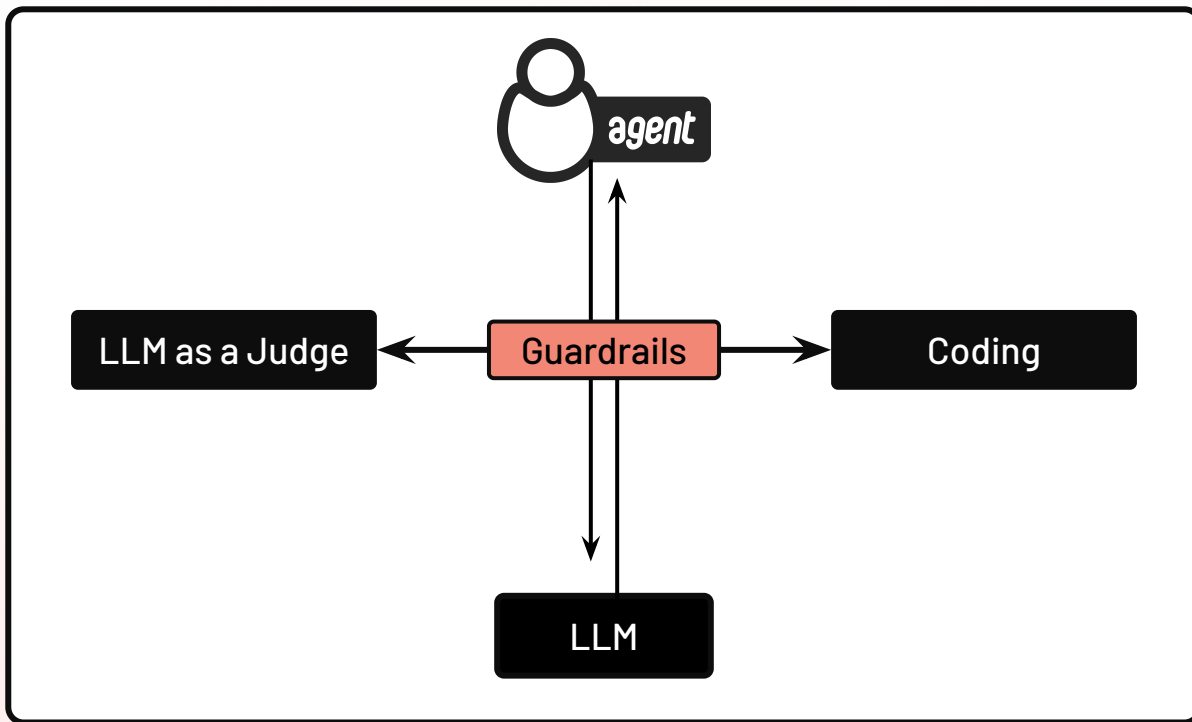
Code Guardrail



Code Guardrail



Guardrails



```
from crewai import LLMGuardrail
```

```
Task(  
    config=self.tasks_config[ 'research_task' ],  
    guardrail=LLMGuardrail(  

```

```
        description="The output must contain exactly 10 items",
```

```
        llm="gpt-4.1"
```

```
)
```

```
)
```

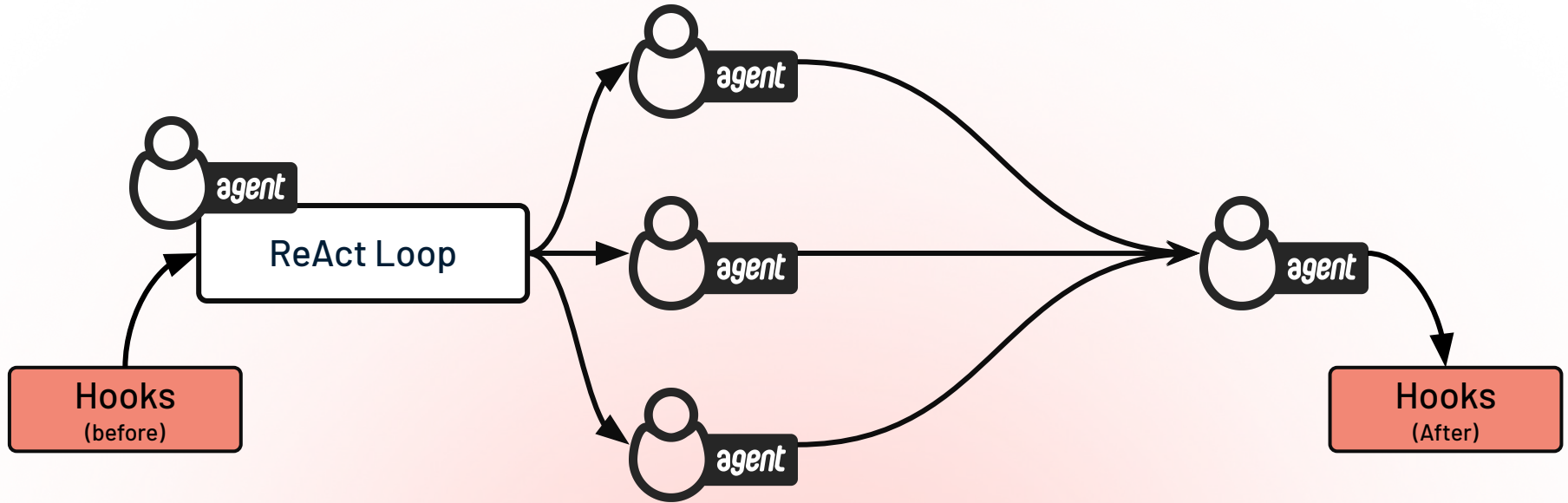
```
def count_of_words(self, result: TaskOutput) -> Tuple[bool, Any]:  
    total_words = len(result.raw.split())  
    if total_words < 500:  
        return (False, "The output must be at least 1000 words.")  
    return (True, "The output is valid.")
```

```
Task(  
    config=self.tasks_config['research_task'],  
    guardrail=self.count_of_words  
)
```


Hooks

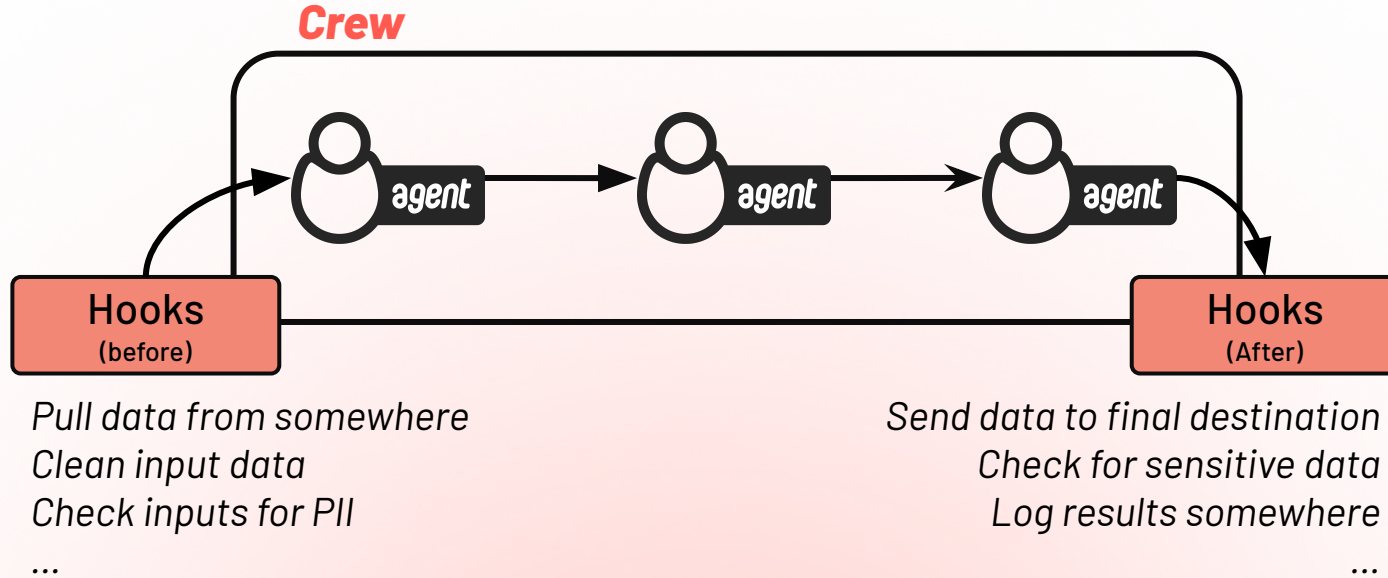
Agent Workflow

(Crews)



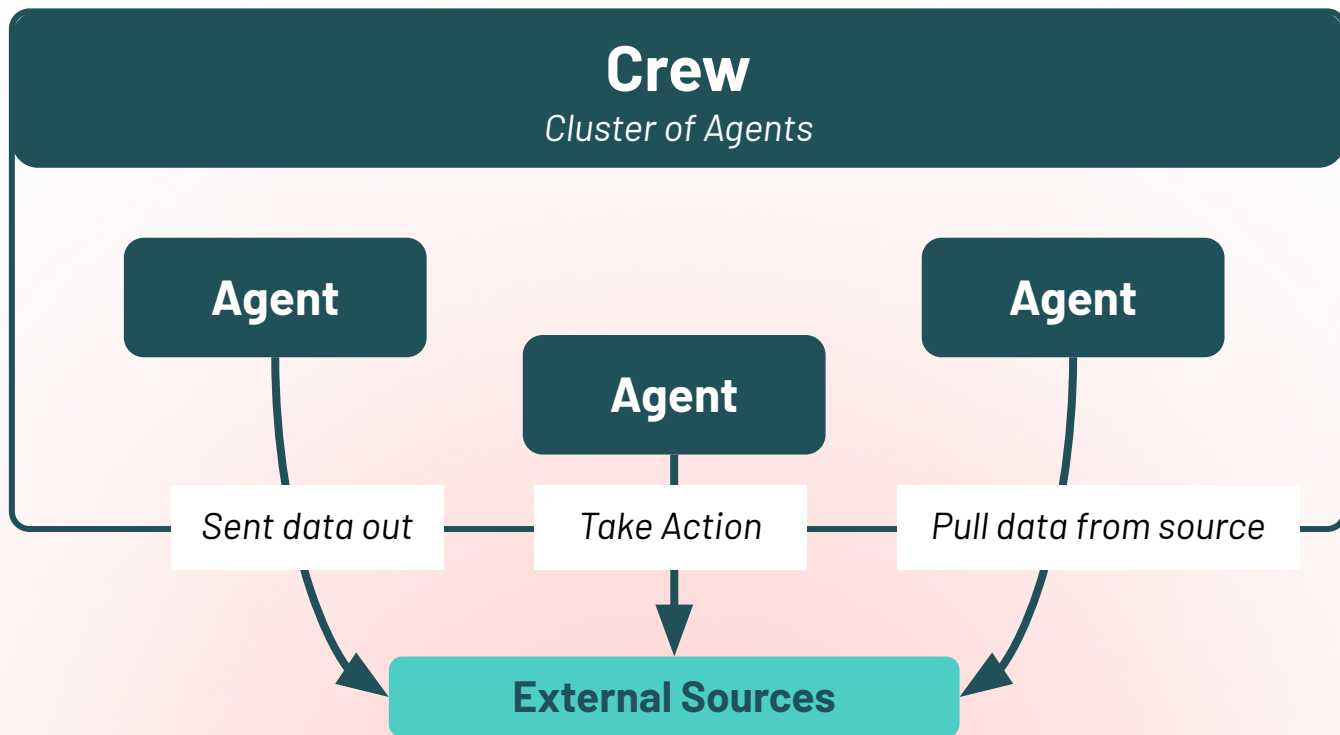
Agent Workflow

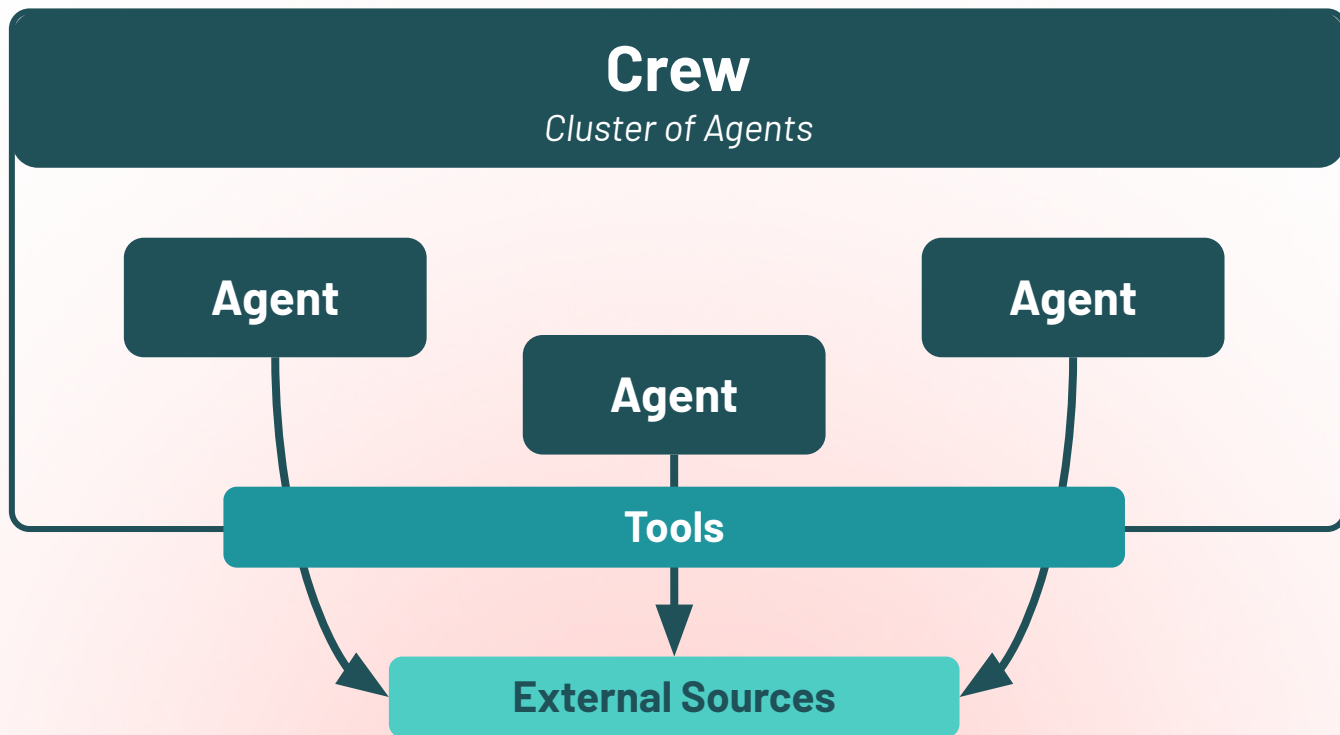
(Crews)

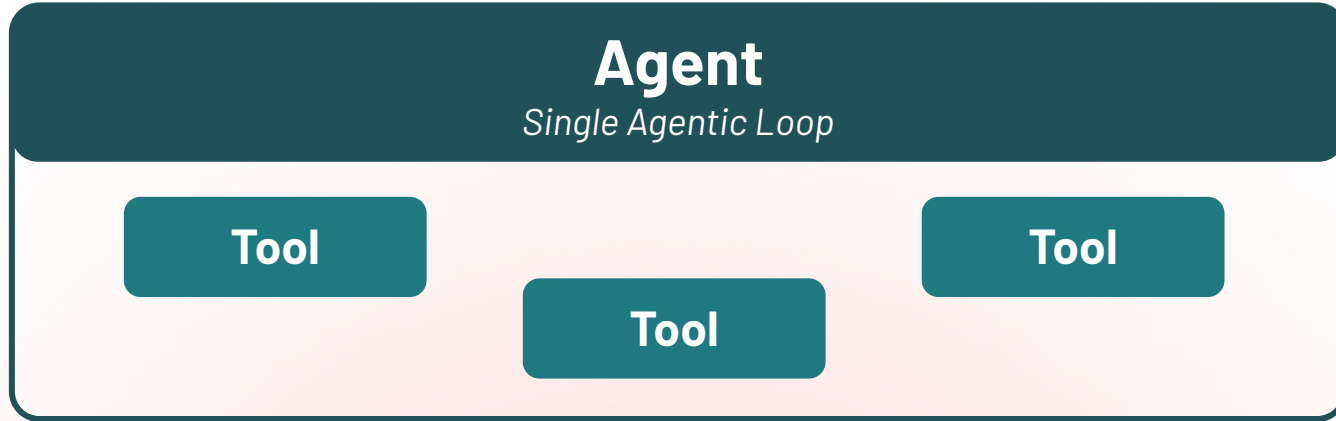


Tools

Let's empower our Agents!







- Tools are extremely lightweight.
- Tools will define what an agent is able to do.
- Giving one agent many tools can be enticing but gets worst results.

Tool

Possible Agent Action

Simple Interface

Custom code



Action Taking Tools

- Write to Database
- Update CRM
- Generate Report

Retrieval Tools

- RAG Services
- External APIs
- Internal Data

Communication Tools

- Email
- Slack
- Teams



Simple Interface

- Name
- Description
- Inputs

Custom Code

- Authentication
- API Calling
- Service Connection



Simple Interface

- Name
- Description
- Inputs

Custom Code

- Authentication
- API Calling
- Service Connection

Scrape Website Tool

Name: *Read website content*

Description: *A tool that can be used to read a website content.*

Inputs: { *website_url: Optional[str]* }

Custom Code

Simple scraping logic using http requests.



Authenticated Integrations



Box

Connect to your Box account to access, create, and update files in Box. Increase your team's productivity by keeping your Box account up to date - without manual data entry. Our Box integration enables...



Gmail

Connect to your Gmail account to manage your emails and drafts in Gmail. Increase your team's productivity by keeping your Gmail account up to date - without manual data entry. Our Gmail integration enables yo...



Google Calendar

Connect your Google account and sync events with your Google Calendar. Increase your productivity by ensuring your schedule is always up to date - without manual data entry. Our Google Calendar...



Confluence

Connect to your Confluence account access, create, and update their documents in Confluence. Increase your team's productivity by keeping your Confluence account up to date - without manual data entry. Our...



Asana

Connect to your users' Asana account to access, create, and update their tasks or projects in Asana. Paragon enables you to sync data between your app and your users' Asana accounts, for example: • Create or...



Salesforce

Connect your Salesforce account and sync your Salesforce accounts, contacts, leads, or opportunities. Enable your sales team to close more deals by keeping your Salesforce CRM records up to date - without...



Zendesk

Connect your Zendesk account to sync tickets from your Zendesk Support account. Enable your support team to deliver better customer experiences by keeping their Zendesk tickets up to date - without manual data entry...



Shopify

Connect your Shopify store and sync your Shopify customers, orders, or products. Grow your business faster by keeping your Shopify store up to date - without manual data entry. Our Shopify integration enables you t...



ClickUp

Connect to your ClickUp account to manage your tasks in ClickUp. Increase your team's productivity by keeping your ClickUp spaces up to date - without manual data entry. Our ClickUp integration enables you t...



Stripe

Connect your Stripe account and sync your Stripe customers, payments, or products. Grow your business faster by keeping your Stripe account up to date - without manual data entry. Our Stripe integration enables you ...



Jira

Connect your Jira Software account to create or update issues in your Jira projects. Increase your team's productivity by keeping your Jira projects and issues up to date - without manual data entry. Our Jira...



Google Sheets

Connect your Google account and sync data with your Google Sheets spreadsheets. Our Google Sheets integration enables you to: • Automatically create or update rows in Google Sheets • Sync...



Microsoft Teams

Connect your Teams workspace to receive notifications and alerts in Teams. Stay connected to important activity by bringing it all together in your Teams workspace. Our Teams integration enables you to: • Receive...



GitHub

Connect to your GitHub account to manage your issues, releases, repositories, and more in GitHub. Increase your team's productivity by keeping your GitHub account up to date - without manual data entry. Our...



HubSpot

Connect your HubSpot account to sync records from your HubSpot CRM. Enable your sales team to close more deals by keeping your HubSpot CRM records up to date - without manual data entry. Our HubSpot...



Google Drive

Connect your Google account and sync files from your Google Drive. Our Google Drive integration enables you to: • Save files to your Google Drive • Sync files from your Google Drive



Linear

Connect to your Linear account to manage your software projects, sprints, tasks, and bug tracking in Linear. Increase your team's productivity by keeping your Linear account up to date - without manual data entry. Our...



Notion

Connect to your Notion account to manage your pages and databases in Notion. Increase your team's productivity by keeping your Notion account up to date - without manual data entry. Our notion integration...



```
from crewai import Agent, Task, Crew
from crewai_tools import CrewaiEnterpriseTools

# Get enterprise tools (Gmail tool will be included)
enterprise_tools = CrewaiEnterpriseTools(
    enterprise_token="your_enterprise_token"
)

# Create an agent with Gmail capabilities
email_agent = Agent(
    role="Email Manager",
    goal="Manage and organize email communications",
    backstory="An AI assistant specialized in email",
    tools=enterprise_tools
)
```

Tool Reliability

- Tool Caching
- Force Tool Return
- Max Usage
- Tool Repository

Tool Repository



Tools

Collaborate by sharing tools within your organization, or publish them publicly to contribute with the community.



Getting Started

1. Creating a new tool? Use `crewai tool create your-tool`
2. Ready to share your tool? Publish with `crewai tool publish`
3. Want to use a tool? Install it with `crewai tool install your-tool`



Your Tools



Youtube Tool Public
Youtube Integration Tool



Databricks Integration Private
Power up your crews with databricks_tool



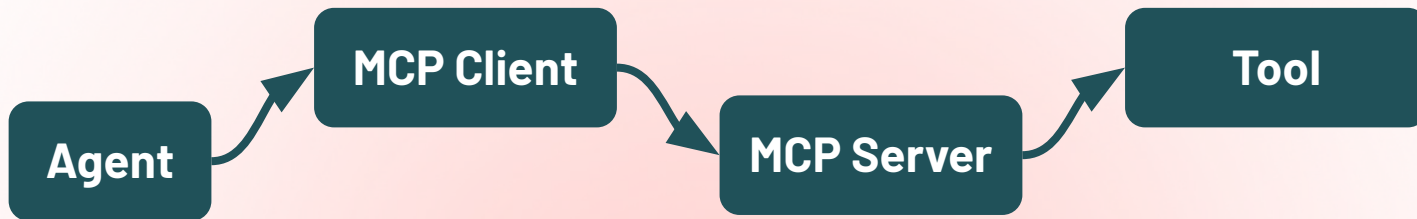
Internal Support System Private
Integration with CrewAI Internal Support System

Before MCP: Fragmentation

- Everyone expose tools differently
- Integrations require bespoke adapters
- Difficult to share tools among agents
- Entire companies built around integrations

What is MCP?

- Open protocol for tools, resources, prompts.
- JSON-RPC 2.0 under the hood.
- Client ↔ Server architecture.



MCP Primitives & Transports

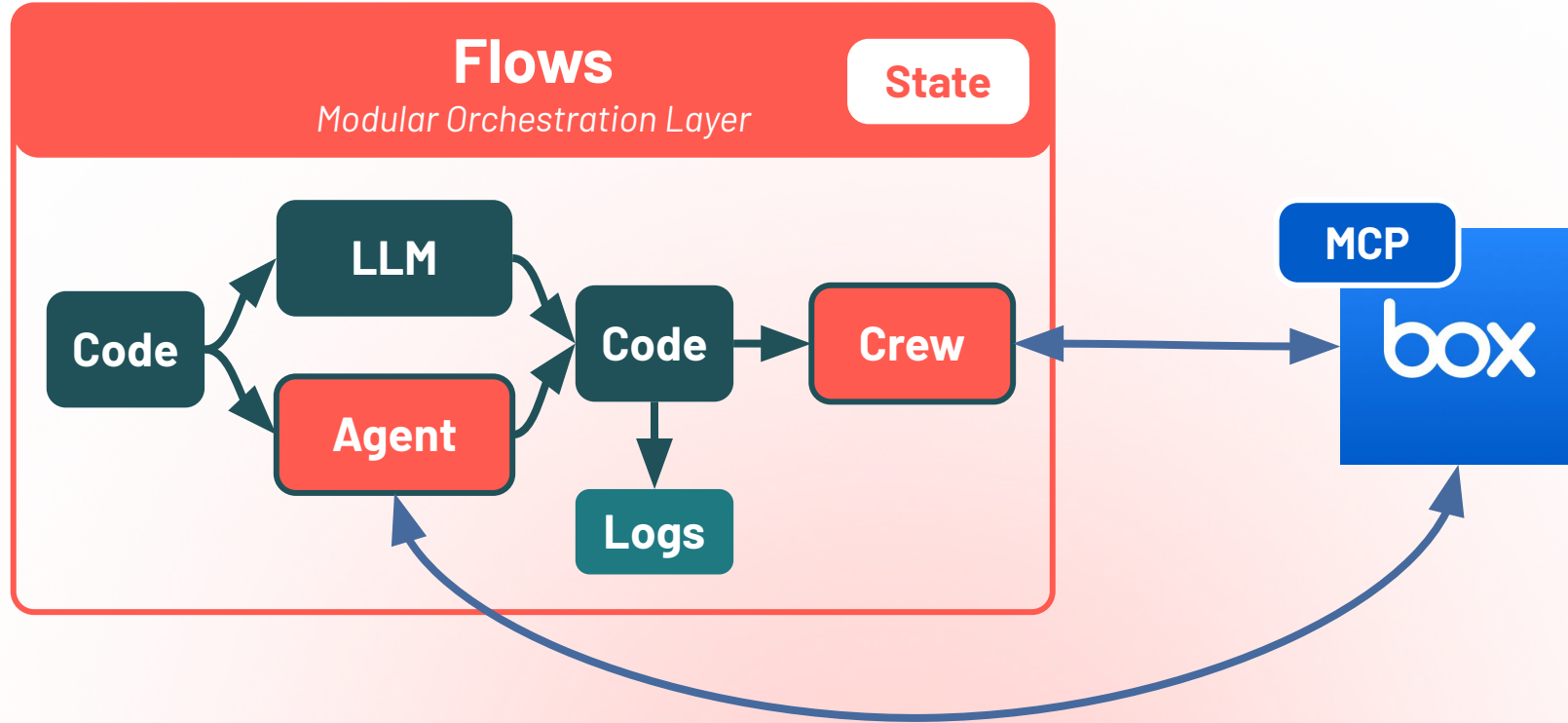
- **Primitives:**

- Tools
- Resources
- Prompts

- **Transports:**

- Stdio (local),
- SSE (streaming),
- Streamable HTTP (remote).

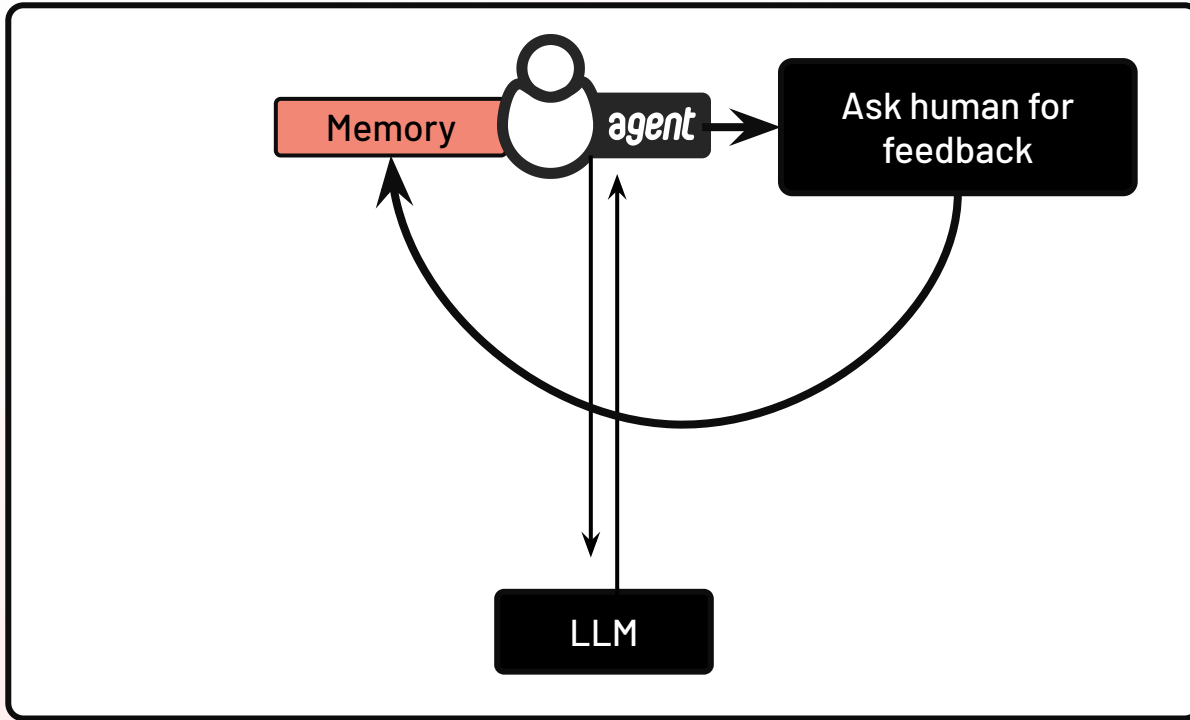
CrewAI & MCP



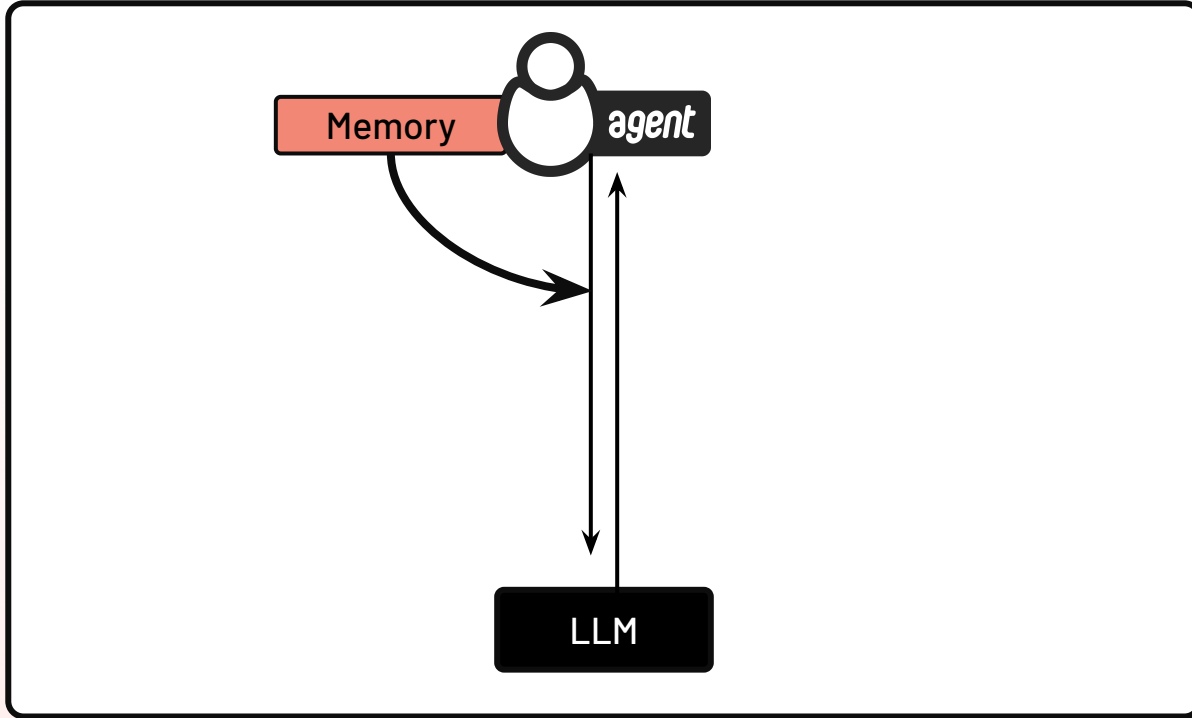
Let's go deeper

...

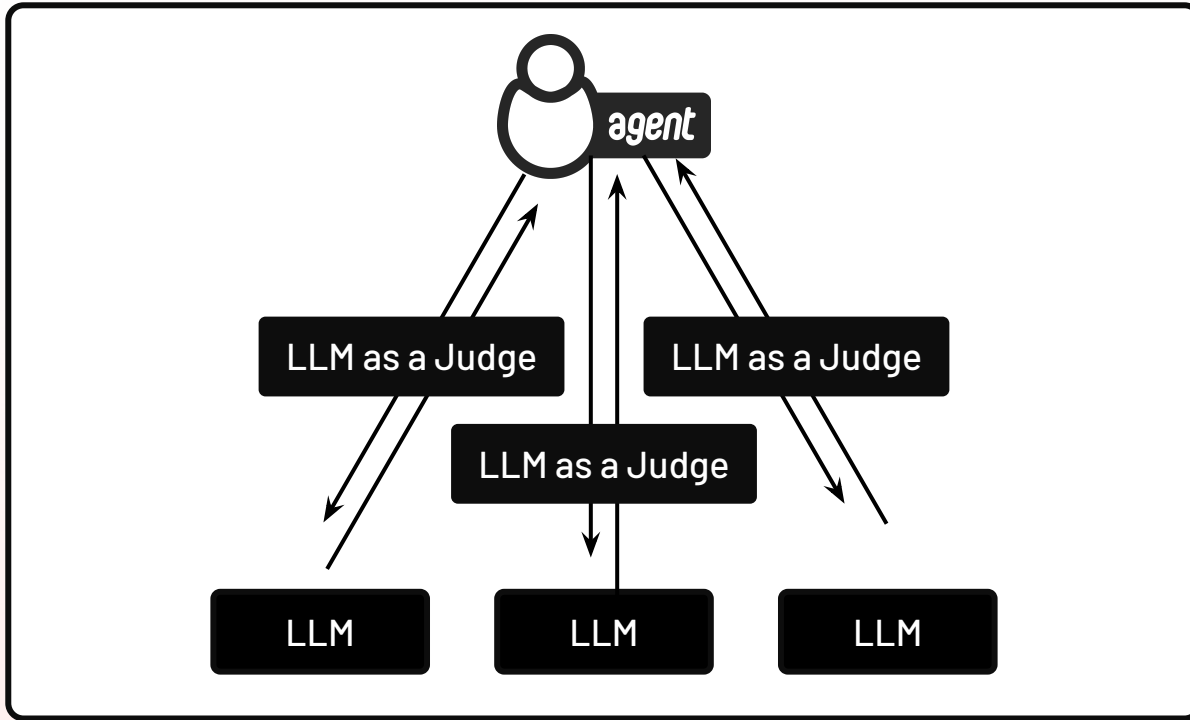
Training



Training

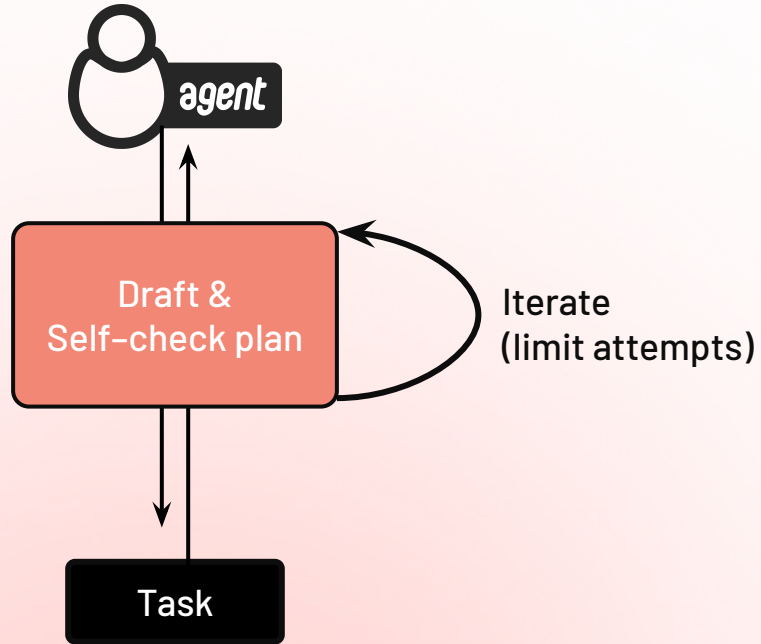


Testing



Turn on Reasoning at the Agent
level to force a pre-plan before
execution

Reasoning



```
from crewai import Agent

# Create an analyst agent with reasoning enabled
analyst = Agent(
    role="Data Analyst",
    goal="Find actionable insights from the data",
    backstory="Experienced at EDA and communication",
    reasoning=True,
    max_reasoning_attempts=3,
    verbose=True
)
```

“You can’t fix what you can’t see.”

```
Crew(  
    agents=self.agents,  
    tasks=self.tasks,  
    process=Process.sequential,  
    verbose=True,  
    tracing=True  
)
```


Crew Quality Sampling

Sampling 10% of executions for quality checks.

Sampling Percentage

Select the portion of executions to be evaluated.



Enable Quality Sampling

Toggle this feature to enable or disable quality sampling for this deployment.

Enabled ☒

▼ Advanced thresholds

Performance Threshold (0-10)

Max Execution Time (seconds)

Tokens, Execution Time and Test Results

Aggregating prompt tokens and execution time. Displaying averaged daily test results.

Time Range

Daily

Period

Current Month

Aggregation

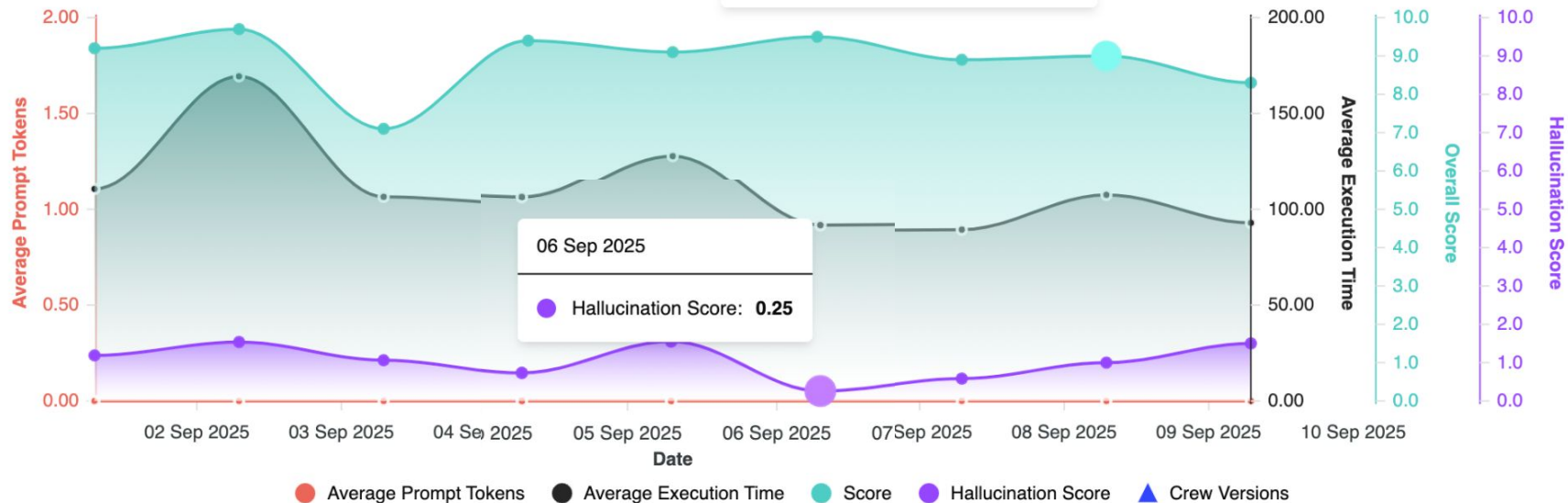
Average

Data type

Discrete

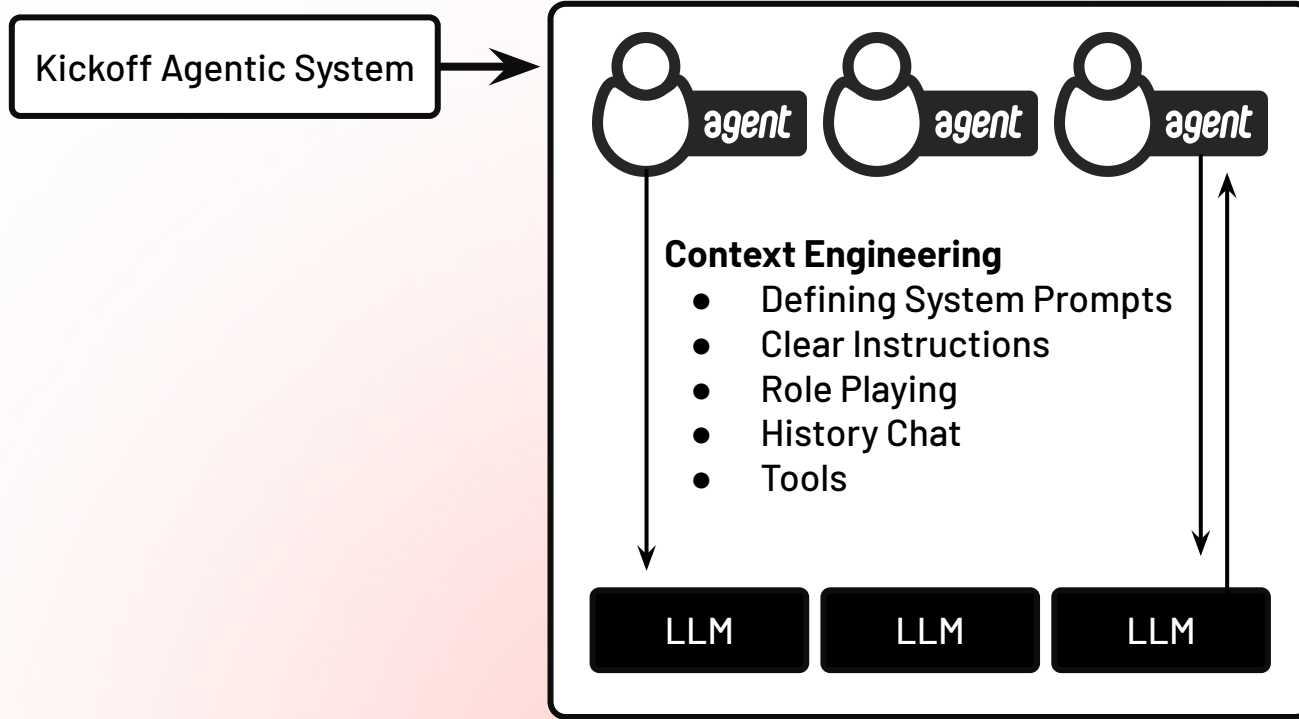
08 Sep 2025

Score: 9 (Hallucination Score: 1.0)

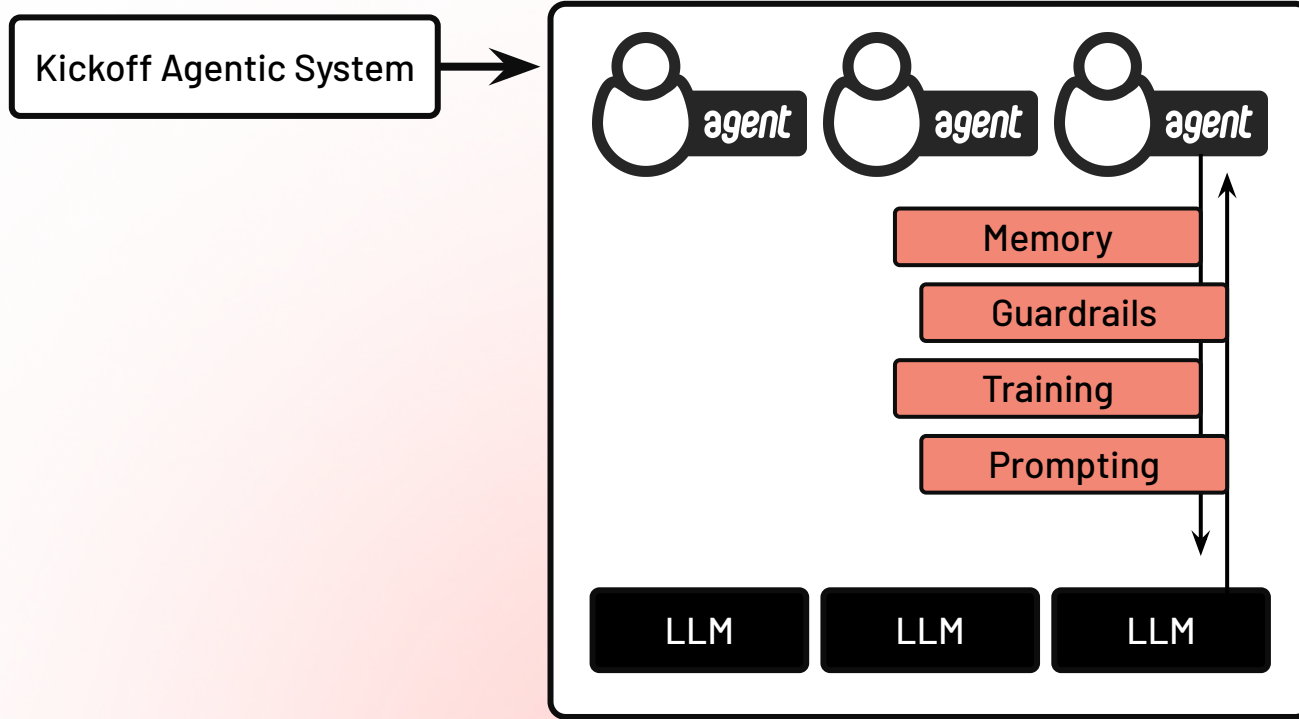


What About Now?

Agentic Systems

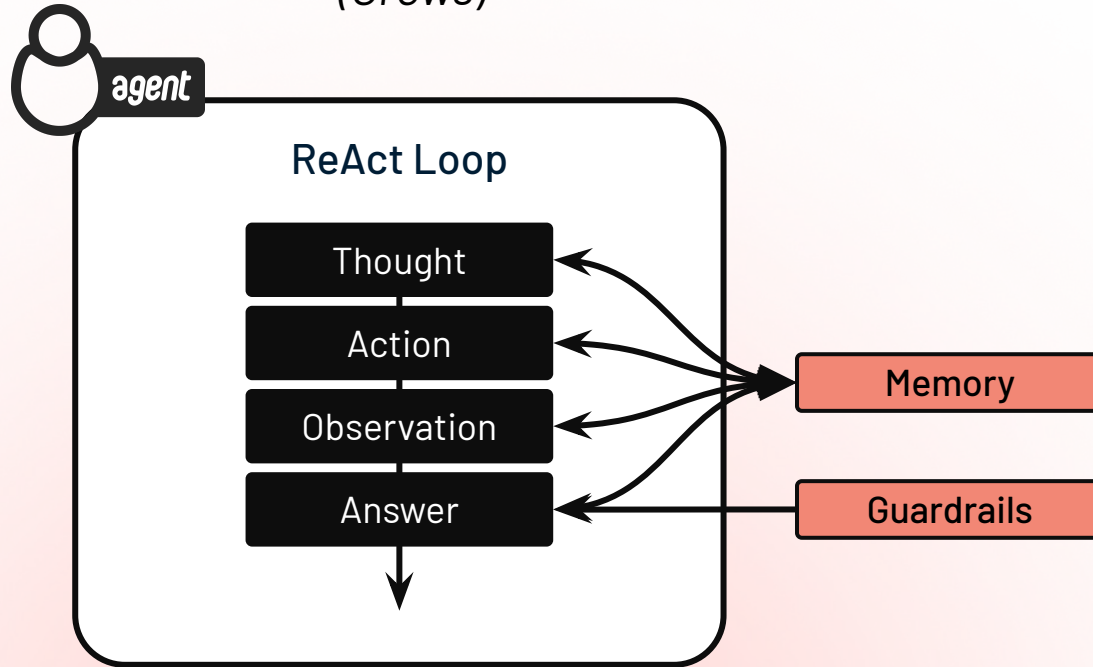


Agentic Systems



Agent Workflow

(Crews)



What about **Pitfalls?**

1. Not spending time on planning use cases
2. Not clear definition of success
3. Not breaking the process into smaller chunks
4. Not measuring / evaluating

Advanced Topics

Structured Output

Enforce Pydantic and Json Output in your
tasks

Event Bus

Use the event bus